DS201 - Final Project
(Dataset : MPG)

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Course Name: Data Science - Coop
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Share notebook : https://ciccc-ds.slack.com/archives/C0AJ2C4GU02/p1772478683347849

[1. Introduction (About the dataset)]

1. **What type of dataset is this?**
    a. It is an **MPG (miles per Gallon) dataset**, containing information about cars and their fuel efficiency, **structured as a tabular dataset** (rows × columns) with a size of **398 rows × 9 columns** containing multiple numerical and categorical variables.

2. **What do the variables represent?**
    a. Mainly a **car performance dataset**, commonly used for regression analysis.

Variable	Meaning
`mpg`	Miles per gallon (fuel efficiency – target variable in many analyses)
`cylinders`	Number of engine cylinders
`displacement`	Engine displacement (size of engine)
`horsepower`	Engine horsepower
`weight`	Vehicle weight
`acceleration`	Time to accelerate (performance metric)
`model_year`	The year the car was manufactured
`origin`	Country/region of origin (categorical)
`name`	Car model name

3. **Are there missing values?**

Yes, only one colum shows missing values:

- `horsepower` → **6 missing values**
- All other columns → **No missing values**

This is a small percentage of missing values (~1.5%)

4. **Any unusual patterns observed?**

Horsepower missing values
Only horsepower has missing data — this is typical in the original MPG dataset.

Mixed data types

- Most variables are numeric.
- `The origin` and `name` are categorical/text.
- `Unnamed: 0` is likely unnecessary (can be dropped).

Wide performance range

- MPG ranges roughly from low-performance (high fuel consumption) cars to more efficient models.
- Weight and displacement likely strongly correlate with MPG (heavier cars → lower MPG).

Model year variation

- Cars span multiple years, which allows trend analysis over time (e.g., fuel efficiency improvements).

[2. Data Exploration]

- Display First 5 rows
- Data shape (rows, columns)
- Data Information - Column name, dtypes
- Summary statistics using
- Count & Percentage of missing values

[2-1. Display First 5 rows]

```
#Display first 5 rows
mpg.head()
```

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin	name
0	18.0	8	307.0	130.0	3504	12.0	70	usa	chevrolet chevelle malibu
1	15.0	8	350.0	165.0	3693	11.5	70	usa	buick skylark 320
2	18.0	8	318.0	150.0	3436	11.0	70	usa	plymouth satellite
3	16.0	8	304.0	150.0	3433	12.0	70	usa	amc rebel sst
4	17.0	8	302.0	140.0	3449	10.5	70	usa	ford torino

[2-2. Data shape (rows, colums)]

```
#Data shape (rows, columns)
rows, columns = mpg.shape
print("Rows:", rows, "/", "Columns:", columns)
```

```
Rows: 398 / Columns: 9
```

[2-3. Data Information]

```
#Data Info - Column name, dtypes,
mpg.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 398 entries, 0 to 397
Data columns (total 9 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   mpg           398 non-null    float64
 1   cylinders     398 non-null    int64
 2   displacement  398 non-null    float64
 3   horsepower    392 non-null    float64
 4   weight        398 non-null    int64
 5   acceleration  398 non-null    float64
 6   model_year    398 non-null    int64
 7   origin        398 non-null    object
 8   name          398 non-null    object
dtypes: float64(4), int64(3), object(2)
memory usage: 28.1+ KB
```

[2-4. Summary statistics using]

```
#Summary statistics using
mpg.describe()
```

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year
count	398.000000	398.000000	398.000000	392.000000	398.000000	398.000000	398.000000
mean	23.514573	5.454774	193.425879	104.469388	2970.424623	15.568090	76.010050
std	7.815984	1.701004	104.269838	38.491160	846.841774	2.757689	3.697627
min	9.000000	3.000000	68.000000	46.000000	1613.000000	8.000000	70.000000
25%	17.500000	4.000000	104.250000	75.000000	2223.750000	13.825000	73.000000
50%	23.000000	4.000000	148.500000	93.500000	2803.500000	15.500000	76.000000
75%	29.000000	8.000000	262.000000	126.000000	3608.000000	17.175000	79.000000
max	46.600000	8.000000	455.000000	230.000000	5140.000000	24.800000	82.000000

[2-5. Count & percentage of missing values]

```
#Count of missing values per column
mpg.isnull().sum()
```

	0
mpg	0
cylinders	0
displacement	0
horsepower	6
weight	0
acceleration	0
model_year	0
origin	0
name	0

dtype: int64

```python
#Percentage of missing values per column
per_miss = round(mpg.isnull().mean() * 100, 2)
per_miss_str = per_miss.astype(str) + '%'
print(per_miss_str)
```

```
mpg              0.0%
cylinders        0.0%
displacement     0.0%
horsepower      1.51%
weight           0.0%
acceleration     0.0%
model_year       0.0%
origin           0.0%
name             0.0%
dtype: object
```

[3. Data Cleaning]

- Check missing values
- Data cleaning to the missing value (median)
- Check the percent of missing value
- Create 2 columns - weight(kg), brand_name

[3-1. Check missing values]

```python
mpg[mpg['horsepower'].isna()]
```

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin	name
32	25.0	4	98.0	NaN	2046	19.0	71	usa	ford pinto
126	21.0	6	200.0	NaN	2875	17.0	74	usa	ford maverick
330	40.9	4	85.0	NaN	1835	17.3	80	europe	renault lecar deluxe
336	23.6	4	140.0	NaN	2905	14.3	80	usa	ford mustang cobra
354	34.5	4	100.0	NaN	2320	15.8	81	europe	renault 18i
374	23.0	4	151.0	NaN	3035	20.5	82	usa	amc concord dl

[3-2. Data Cleaning - missing value (median)]

```python
mpg['horsepower'].median()
```

=> 93.5

```
#copy to new dataset
mpg_new = mpg.copy()
mpg_new['horsepower'] = mpg['horsepower'].fillna(mpg['horsepower'].median())
#mpg_new.head()
mpg_new.iloc[[32, 126, 330, 336, 354, 374]]

#mpg_new.isnull().sum()
```

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin	name
32	25.0	4	98.0	93.5	2046	19.0	71	usa	ford pinto
126	21.0	6	200.0	93.5	2875	17.0	74	usa	ford maverick
330	40.9	4	85.0	93.5	1835	17.3	80	europe	renault lecar deluxe
336	23.6	4	140.0	93.5	2905	14.3	80	usa	ford mustang cobra
354	34.5	4	100.0	93.5	2320	15.8	81	europe	renault 18i
374	23.0	4	151.0	93.5	3035	20.5	82	usa	amc concord dl

[3-3. Check the percent of missing value]

```
check_miss = mpg_new.isna().mean()
check_miss_str = check_miss.astype(str) + '%'
print(check_miss_str)
```

```
mpg             0.0%
cylinders       0.0%
displacement    0.0%
horsepower      0.0%
weight          0.0%
acceleration    0.0%
model_year      0.0%
origin          0.0%
name            0.0%
dtype: object
```

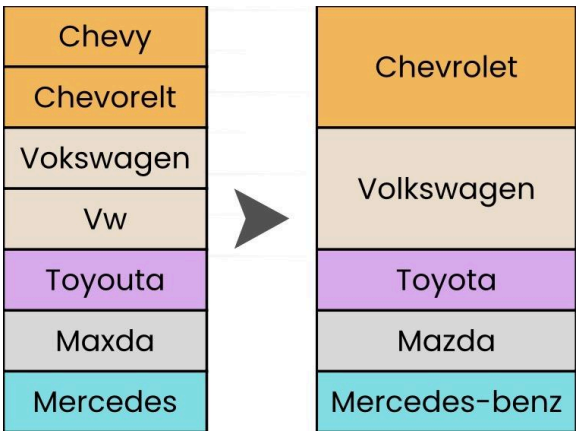
[3-4. Create 2 columns - weight(kg), brand_name]

- weight = weight / 1000
- brand_name : extract the first string of the name

```
#Create new column
# mpg_new['weight(kg)']
mpg_new['weight(kg)'] = round(mpg_new['weight'] / 1000, 2)

#mpg_new['brand_name']
mpg_new['brand_name'] = mpg_new['name'].str.split(' ').str[0].str.capitalize()
```

- Cleaned up the brand_name




```
brand_map = {
    "Chevroelt": "Chevrolet",
    "Chevy": "Chevrolet",
    "Vokswagen": "Volkswagen",
    "Vw": "Volkswagen",
    "Toyouta": "Toyota",
    "Maxda": "Mazda",
    "Mercedes" : "Mercedes-benz"
}

mpg_new['brand_name'] = mpg_new['brand_name'].replace(brand_map)
```

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin	name	weight(kg)	brand_name
0	18.0	8	307.0	130.0	3504	12.0	70	usa	chevrolet chevelle malibu	3.50	Chevrolet
1	15.0	8	350.0	165.0	3693	11.5	70	usa	buick skylark 320	3.69	Buick
2	18.0	8	318.0	150.0	3436	11.0	70	usa	plymouth satellite	3.44	Plymouth
3	16.0	8	304.0	150.0	3433	12.0	70	usa	amc rebel sst	3.43	Amc
4	17.0	8	302.0	140.0	3449	10.5	70	usa	ford torino	3.45	Ford

[4. Data Visualization]

Part 3: Data Visualization (Using Matplotlib ONLY)

[Chart List]

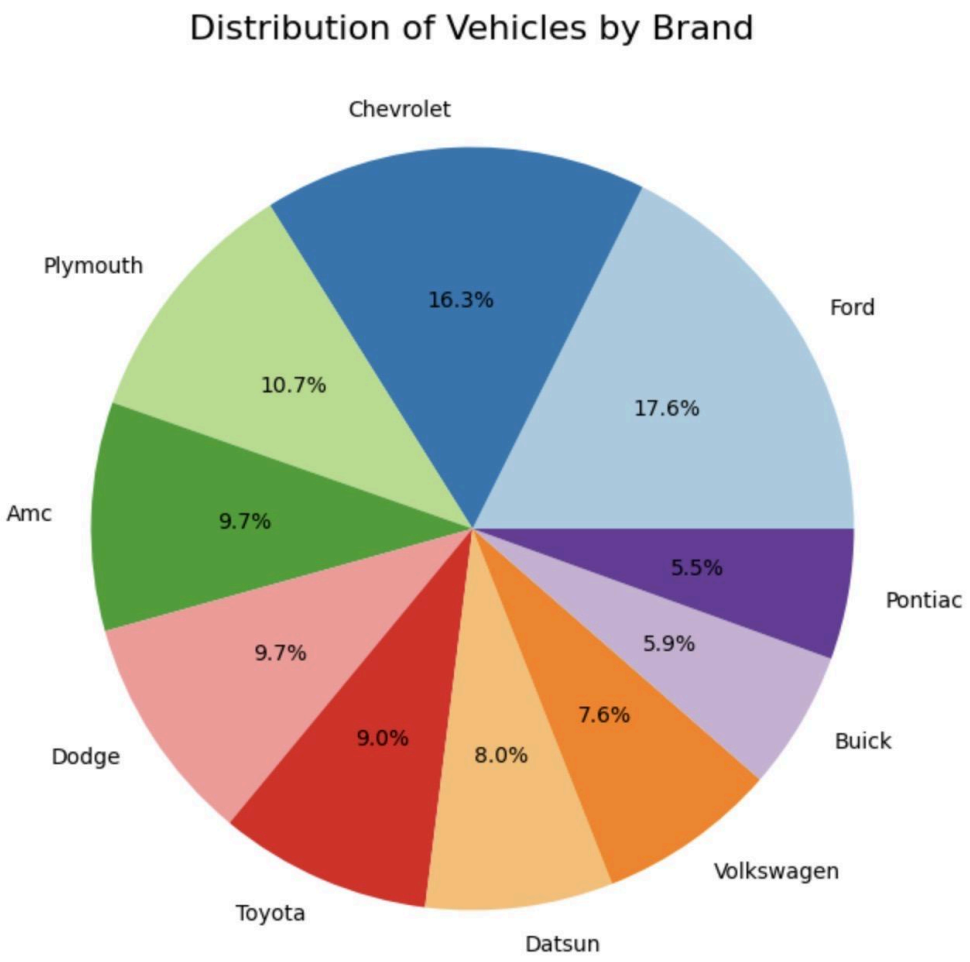
- Distribution of Vehicles by Brand
- Top 10 Most Efficient Vehicles (Highest MPG)
- Top 10 Less Efficient Vehicles (Lowest MPG)
- Vehicle Weight(KG) VS Fuel Efficiency
- Average Fuel Efficiency(MPG) Trend by Region (1972-1980)
- Evolution of Fuel Efficiency: Average MPG per Year (1970-1982)

[4-1 Distribution of Vehicles by Brands]

This graph quantifies the composition of the dataset. It answers the question: "How much representation does each manufacturer have in this study?" By measuring the count of each brand, we identify which manufacturers are the primary influencers of the overall averages.

- **What it measures:** Frequency of occurrence (Categorical Volume).
- **Variables:**
    - X-Axis: brand_name. This is a qualitative, nominal variable.
    - Y-Axis: Count. The number of vehicles assigned to that brand.

This chart shows the market share of each brand in the dataset.

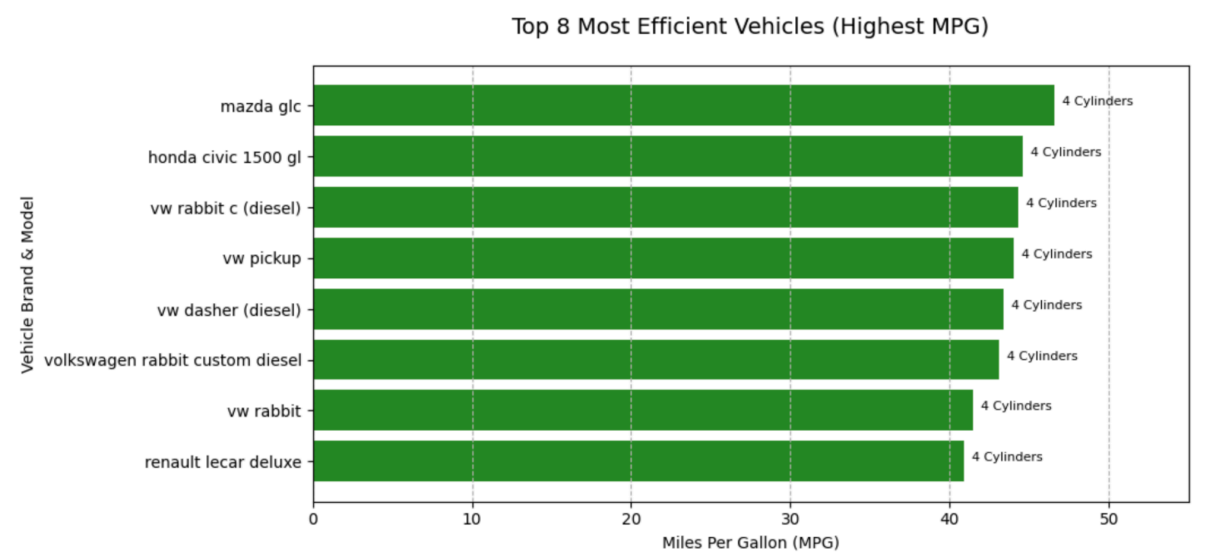


[4-2. Top 8 Most Efficient Vehicles (Highest MPG)]

This is a ranking chart. It measures the maximum distance a vehicle can travel per gallon of fuel. By including the cylinder count, it also measures the technical commonality among the "winners" (identifying that high MPG is tied to low cylinder counts).

- **What it measures:** Peak Performance (Ranking of Extremes).
- **Variables:**
    - X-Axis: brand_name + name (Vehicle identity).
    - Y-Axis: mpg (Miles Per Gallon). This is a continuous quantitative variable representing fuel economy.
    - cylinders. A quantitative variable used as a label to show the engine type of these leaders.

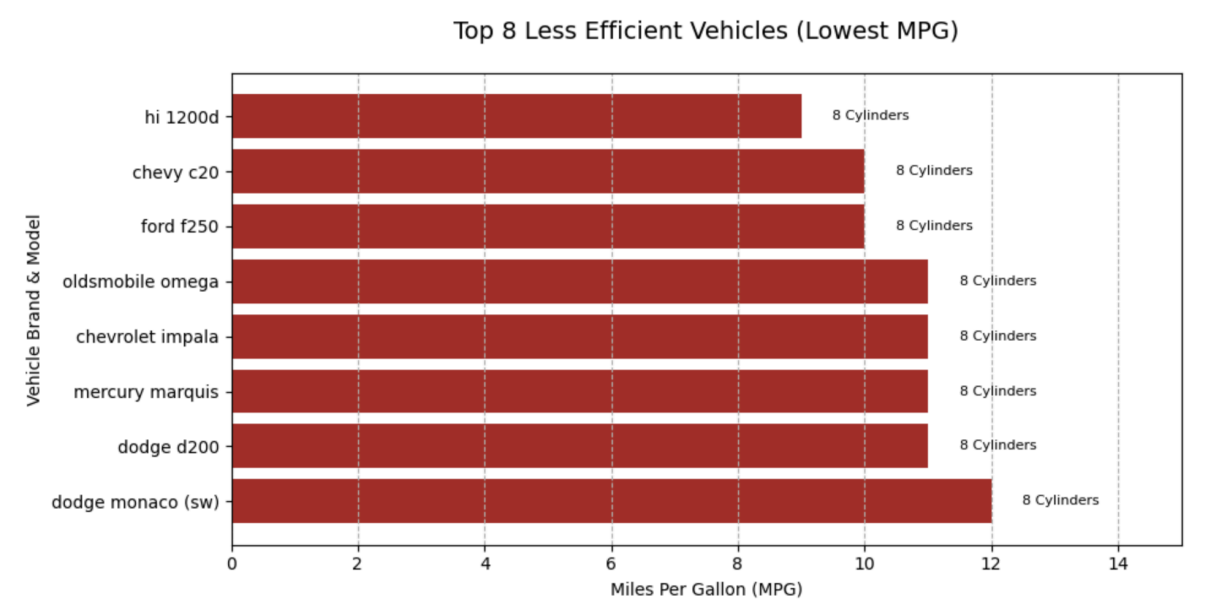
This horizontal bar chart displays the 8 car models in the dataset that achieve the highest fuel efficiency (MPG).



[4-3. Top 8 Less Efficient Vehicles (Lowest MPG)]

Similar to the previous chart, but measuring the opposite end of the spectrum. It identifies the "worst performers." It measures how much fuel is consumed to move these vehicles, highlighting the correlation between high cylinder counts (8 cylinders) and low efficiency.

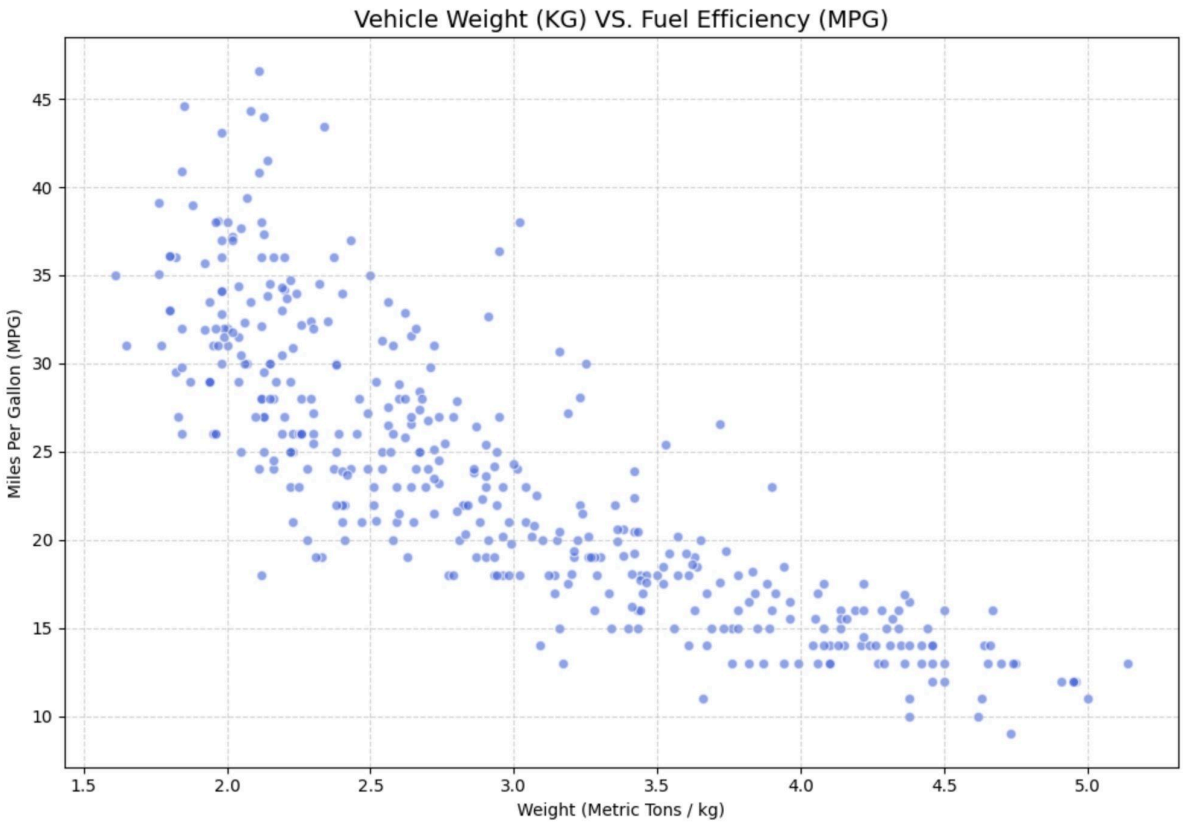
- **What it measures:** Performance Deficit (Ranking of Extremes).
- **Variables:**
    - X-Axis: brand_name + name.
    - Y-Axis: mpg (Miles Per Gallon).
    - cylinders.



[4-4. Vehicle Weight(KG) VS Fuel Efficiency]

This graph measures the direct physical relationship between two variables. Each point represents one vehicle's "Weight-to-Economy" ratio. It is designed to show how mass affects energy consumption—specifically, how an increase in one variable (weight) causes a decrease in the other (MPG).

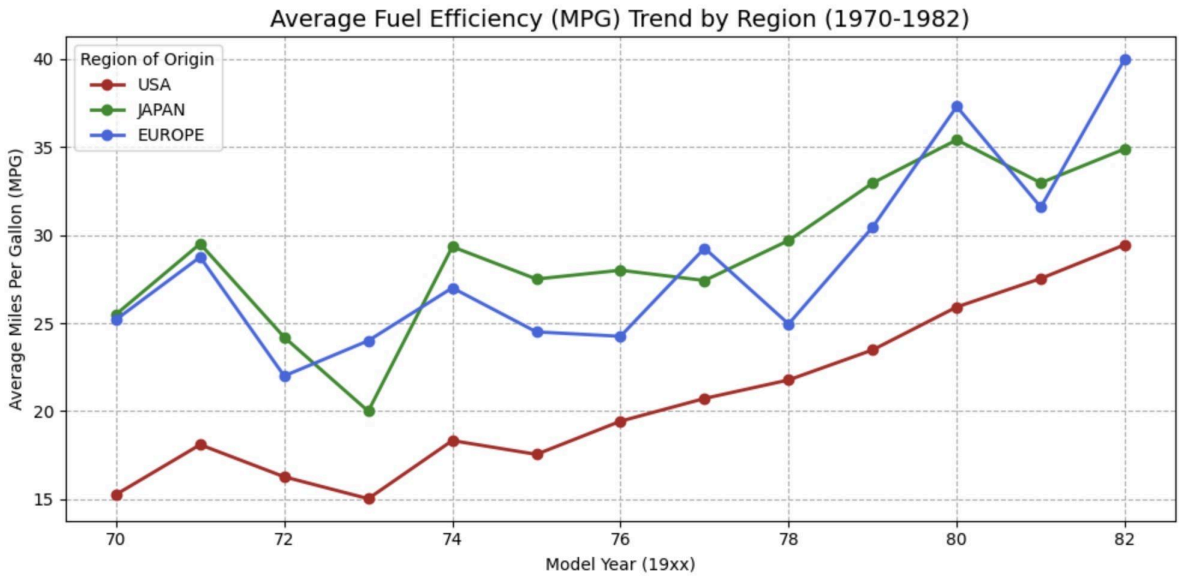
- **What it measures:** Statistical Correlation (Bivariate Analysis).
- **Variables:**
  - X-Axis: weight(kg). Measures the mass of the vehicle.
  - Y-Axis: mpg. Measures fuel economy.



[4-5. Average Fuel Efficiency(MPG) Trend by Region (1970-1982)]

This chart measures the rate of progress across different geographic regions. It doesn't just show that cars got better; it measures *who* got better *faster*. It compares the "efficiency gap" between regions over a 12-year timeline.

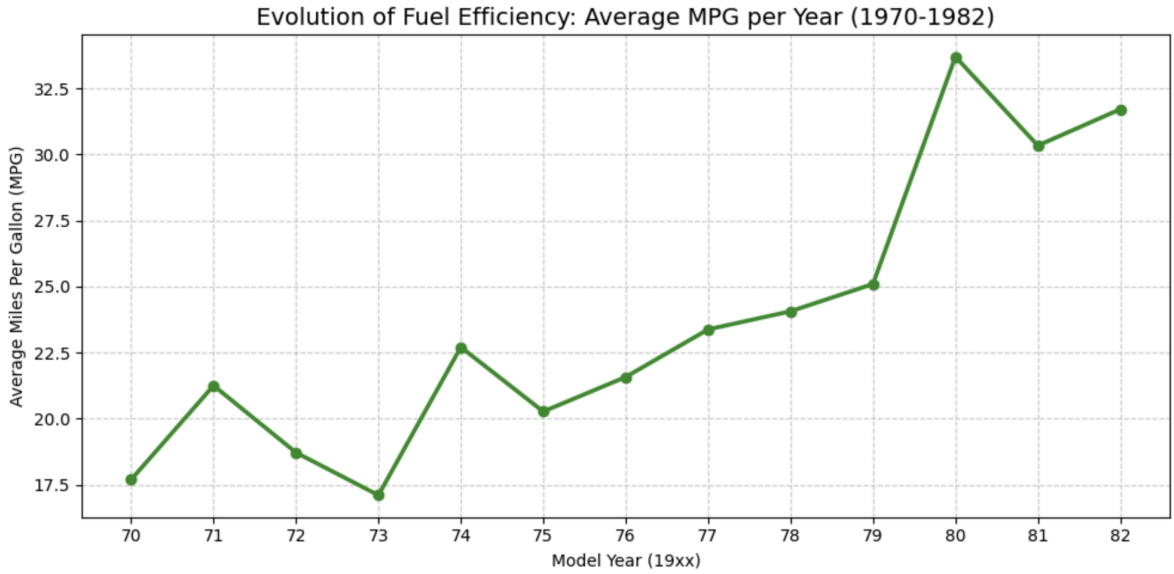
- **What it measures:** Temporal Evolution by Category
- **Variables:**
  - X-Axis: model_year. Quantitative variable representing time.
  - Y-Axis: Mean mpg. The average fuel economy calculated for each group.
  - Grouping Variable: origin. A categorical variable (USA, Japan, Europe) used to create separate lines.



[4-6. Evolution of Fuel Efficiency: Average MPG per Year (1970-1982)]

This graph measures the "Macro Trend" of the entire automotive market. It ignores individual brands and regions to measure the collective technological advancement of the industry. It calculates the arithmetic mean of all vehicles produced in a single year to show the general direction of automotive engineering.

- **What it measures:** Global Industry Progress.
- **Variables:**
  - X-Axis: model_year.
  - Y-Axis: Annual Mean mpg.



[5. Key Insights (3–5)]

1. Ford and Chevrolet make up the dataset

- Ford (~17.6%) and Chevrolet (~16.3%) together form about one-third of all vehicles.
- This suggests that the dataset is severely weighted toward major manufacturers of the USA.

2. Leaders of fuel efficiency are extremely small 4-cylinder cars

- Every vehicle in the top-10 MPG list makes use of **4 cylinders**.
- This strongly supports the expected relationship: fewer cylinders leads to better fuel economy.

3. Import brands (especially Japanese ones and European ones) lead MPG rankings

- Brands such as Mazda, Honda, Volkswagen, Datsun, and Renault make up the efficiency list.
- Despite the brands of the U.S. being common overall, they are largely missing from the top MPG performers.

4. Volkswagen models show up repeatedly among efficient vehicles

- Multiple VW models (Rabbit, Dasher, Pickup, etc.) appear in the top-10.
- This suggests Volkswagen had particularly high fuel-efficiency performance in the dataset.

5. Brand distribution is adequately spread beyond the top two

- After Ford and Chevrolet, most of the brands cluster between ~5–11%.
- No single brand (beyond the top two) extremely leads the remaining share.

[5-1 Patterns Discovered]

Pattern 1: Clear relationship between efficiency and engine size

The consistent presence of 4-cylinder engines among top MPG vehicles displays a strong structural pattern in the data.

Pattern 2: Gap between domestic vs. import

- Dataset composition leads to being dominated by brands of the U.S.
- Efficiency leaderboard leads to being dominated by import brands
  This indicates a consequential trade-off pattern between market presence and fuel efficiency in this dataset.

Pattern 3: Bias of compact car in high MPG

- Models listed (such as Civic, Rabbit, Mazda GLC.) are all compact/subcompact cars.
- Larger cars are definitely missing from the high-MPG range.

Pattern 4: Efficiency of Volkswagen cluster

- VW showing up multiple times indicates a manufacturer-specific strength rather than random distribution.

[5.2 Conclusions Drawn]

1. **Brand Distribution:**

   Ford and Chevrolet are dominants in the dataset, so that indicates the large market share of these two brands during the period.

2. **Fuel Efficiency:**

   A. The top 10 most efficient vehicles are 4 cylinder cars from Japan or Europe, while the top 10 less efficient vehicles are all American cars with 8 cylinders. This can entail the fact that in this era, the price of gasoline in Japan or Europe might be higher, so people need to find cars that allow them to save on more gasoline. On the other hand, gasoline in North America might be cheaper so most people care less about fuel efficiency.

B. The chart “Average Fuel Efficiency (MPG) Trend by Region (1970-1982)” indicates that all cars—-whether they are Japanese, European, or American, all had performance enhancements. This might be related to the oil crisis in the 70s, especially in 1973 and 1979, the wars impacted strongly on the price of oil.

C. The chart “Evolution of Fuel Efficiency: Average MPG per Year (1970-1982) shows the progression of MPG throughout the years. This tells us that if we buy cars that are manufactured later, there is more possibility that the cars are fuel efficient.

3. **Impact of Weight on MPG:**

   A strong negative correlation shows that the heavier the car weight, the more fuel it consumes. This entails that if someone wants to save more on gasoline, they should consider the weight and avoid heavy vehicles if possible.

[5.3 If this were a business dataset, what recommendations would you give?]

1. There is a clear market for fuel-efficient cars (Japanese and European). Investing in R&D for lighter materials or 4-cylinders engines could be important, especially when targeting markets where fuel prices are higher, or consumers with high environmental concerns.

2. The market in North America is leaning more towards heavier cars. This suggests that investing in R&D to reduce weight or make hybrid solutions without sacrifice their size is probably a good idea. Keep the cars with 8 cylinders or bigger size, and only find ways to reduce the use of gasoline to still conform with the market’s purchasing habits.

3. Develop cars that are inherently more fuel-efficient. Anticipating future regulations and consumer demands for lower operating costs. This can also be related to the sales department. When advising car buyers, talk about the future trends of the prices rising on gasoline, and government regulations to encourage them buying fuel-efficient cars even when it’s more pricey when purchasing.